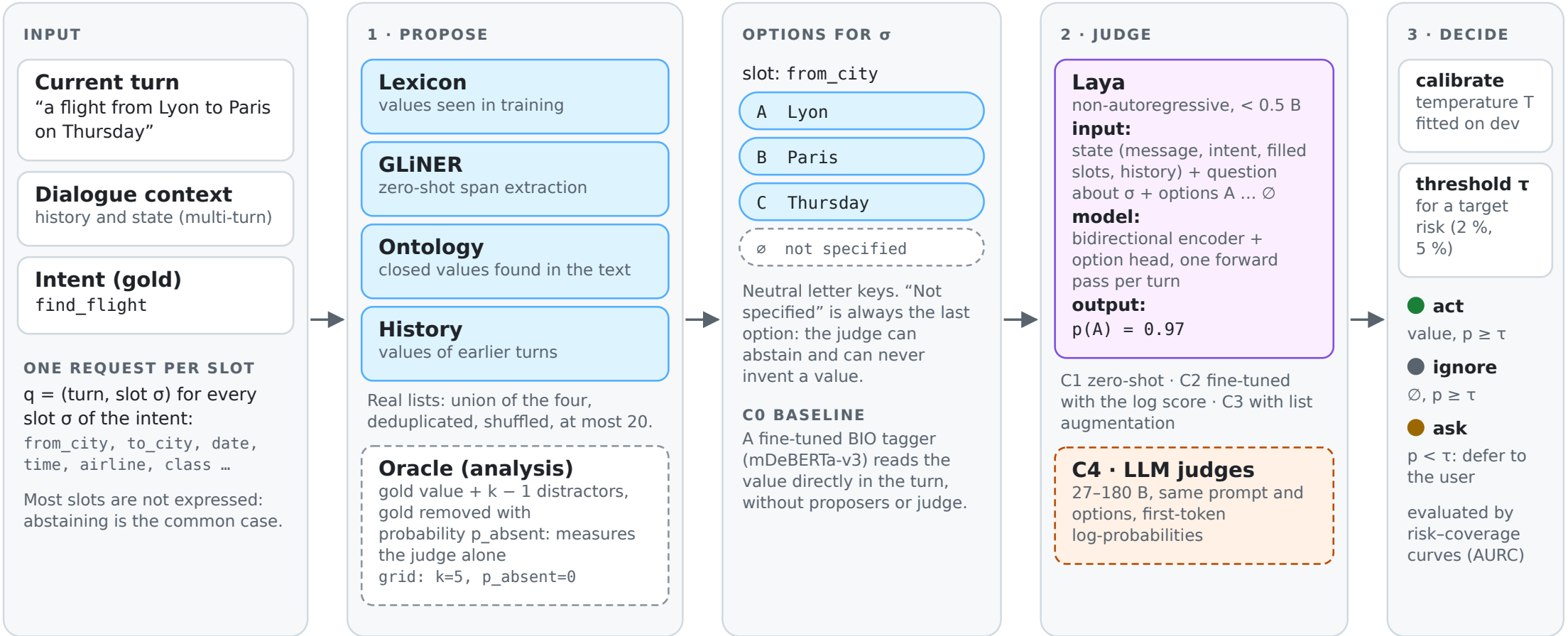

INPUT
Current turn
“a flight from Lyon to Paris on Thursday”
Dialogue context
history and state (multi-turn)
Intent (gold)
find_flight
ONE REQUEST PER SLOT
q = (turn, slot σ) for every slot σ of the intent:
from_city, to_city, date, time, airline, class …
Most slots are not expressed: abstaining is the common case.
1 · PROPOSE
Lexicon
values seen in training
GLiNER
zero-shot span extraction
Ontology
closed values found in the text
History
values of earlier turns
Real lists: union of the four, deduplicated, shuffled, at most 20.
Oracle (analysis)
gold value + k − 1 distractors, gold removed with probability p_absent: measures the judge alone
grid: k=5, p_absent=0
OPTIONS FOR σ
slot: from_city
A Lyon
B Paris
C Thursday
∅ not specified
Neutral letter keys. “Not specified” is always the last option: the judge can abstain and can never invent a value.
C0 BASELINE
A fine-tuned BIO tagger (mDeBERTa-v3) reads the value directly in the turn, without proposers or judge.
2 · JUDGE
Laya
non-autoregressive, < 0.5 B
input:
state (message, intent, filled slots, history) + question about σ + options A … ∅
model:
bidirectional encoder + option head, one forward pass per turn
output:
p(A) = 0.97
C1 zero-shot · C2 fine-tuned with the log score · C3 with list augmentation
C4 · LLM judges
27–180 B, same prompt and options, first-token log-probabilities
3 · DECIDE
calibrate
temperature T fitted on dev
threshold τ
for a target risk (2 %, 5 %)
act
value, p ≥ τ
ignore
∅, p ≥ τ
ask
p < τ: defer to the user
evaluated by risk–coverage curves (AURC)